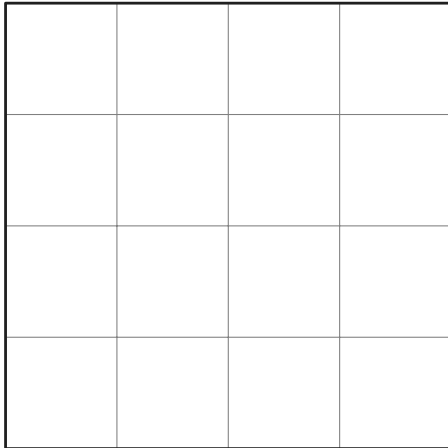
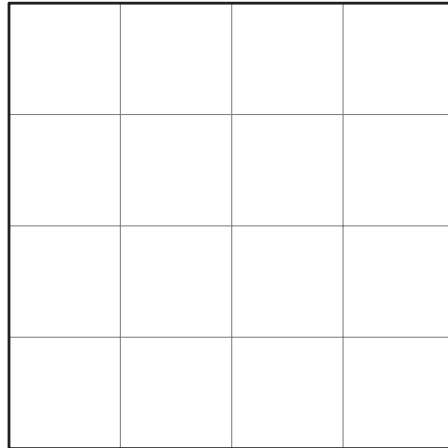
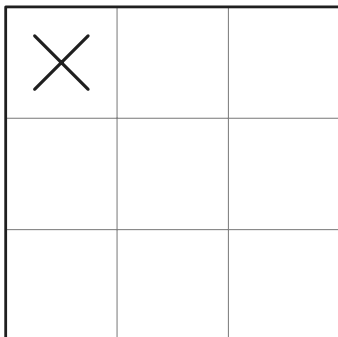


**Problem 1:** A domino is a rectangle that covers exactly two squares sharing a side. Cover board A with 8 paper dominoes so that every square is covered exactly once and nothing sticks out past the edge. Then draw your dominoes on the board: outline each domino. Cover board B in a different way and draw it. Count the vertical dominoes in each drawing and write the number in the box under the board.

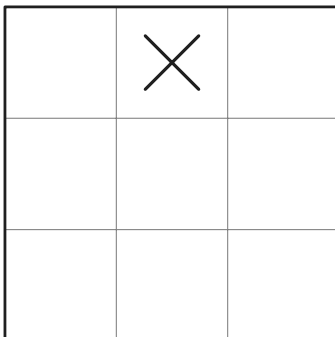
**A****B**

**Problem 2:** Each board below is a 3-by-3 board with one square crossed out. Cover the 8 remaining squares with 4 paper dominoes if you can, and draw the dominoes. If you cannot, write  $\times$  in the box under the board.

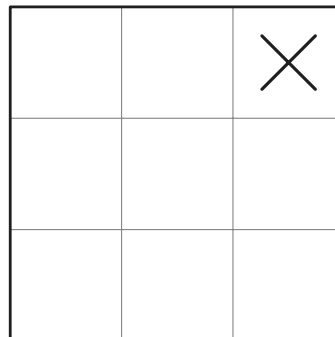
(a)



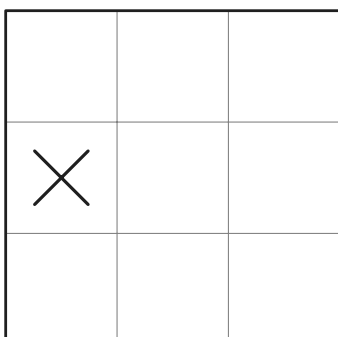

(b)



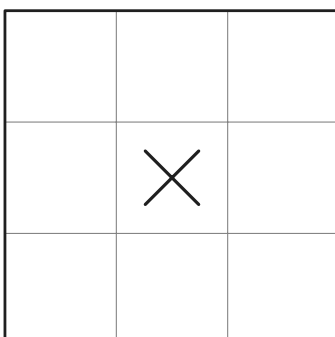

(c)



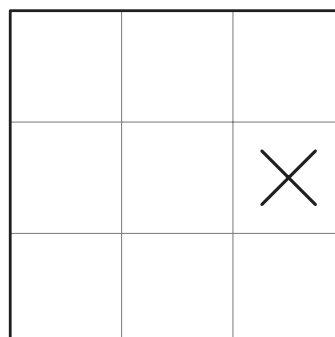

(d)



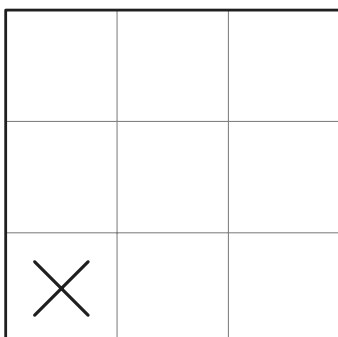

(e)



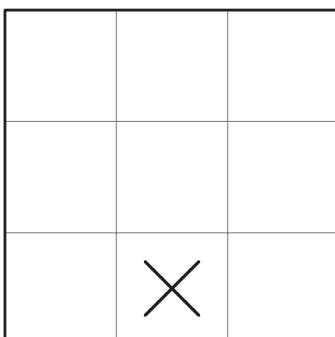

(f)



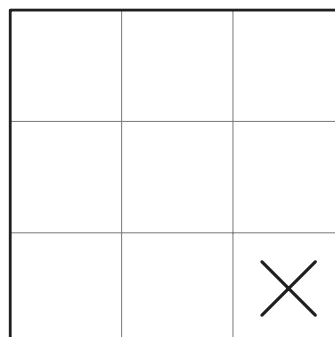

(g)



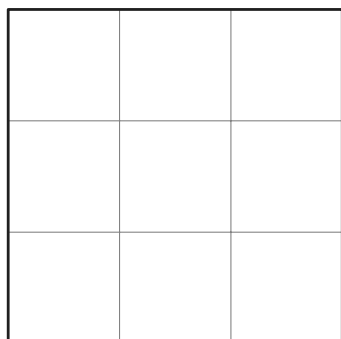

(h)




(i)



**Problem 3:** Color the 3-by-3 board below like a checkerboard: the top-left square dark, and no two squares that share a side the same color. Write how many dark squares and how many light squares there are. Then put one domino anywhere on the colored board and write in the table how many dark and how many light squares it covers. Do this for three different positions.



dark squares:

light squares:

| domino     | dark | light |
|------------|------|-------|
| position 1 |      |       |
| position 2 |      |       |
| position 3 |      |       |

**Problem 4:** Explain why the boards you marked  $\times$  in Problem 2 cannot be covered. Use the counts from Problem 3.

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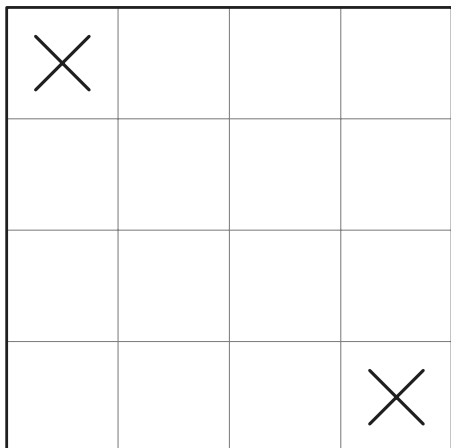
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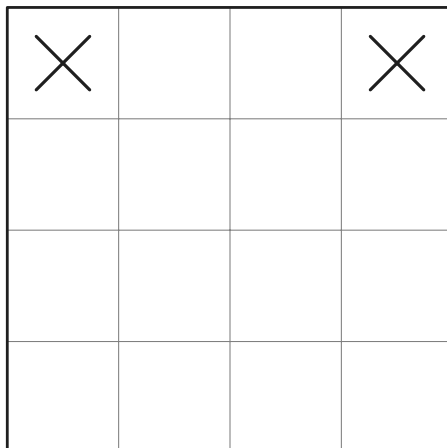
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**Problem 5:** Each 4-by-4 board below has two squares crossed out. For each board, before touching any pieces, write in the table how many dark and how many light squares are left (color the board like a checkerboard if you need to) and your prediction, ✓ or ×. Then check with dominoes and write what happened.

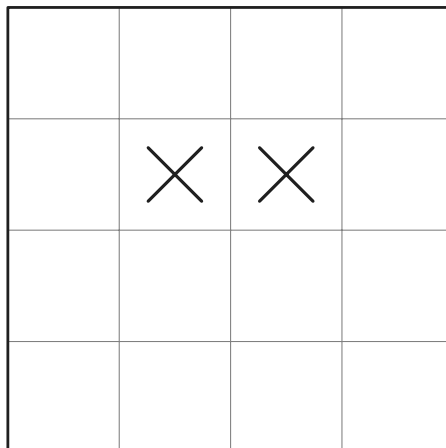
(a)



(b)

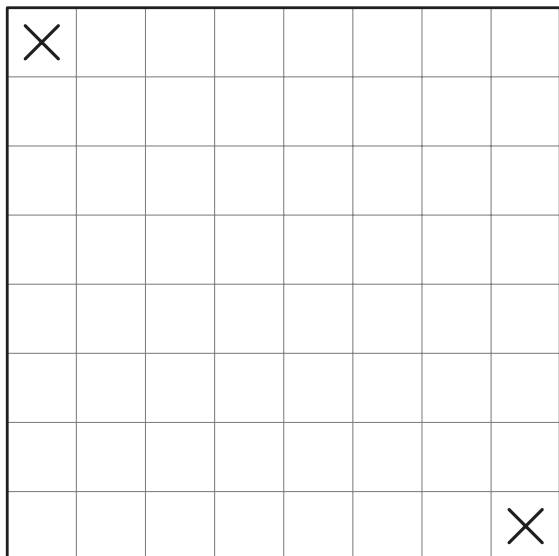
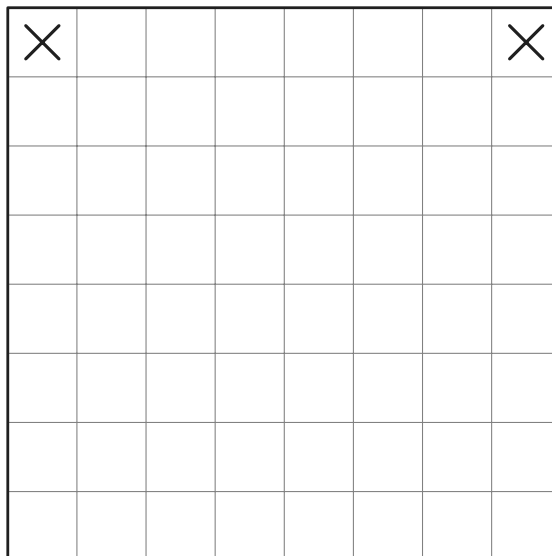


(c)



| board | dark left | light left | prediction | result |
|-------|-----------|------------|------------|--------|
| (a)   |           |            |            |        |
| (b)   |           |            |            |        |
| (c)   |           |            |            |        |

**Problem 6:** Board T is an 8-by-8 board with two opposite corners crossed out. Board U has two corners on the same side crossed out. Without using pieces, write ✓ or × for each board and give your reason on the lines. For any board you marked ✓, draw a covering.

**T****U**


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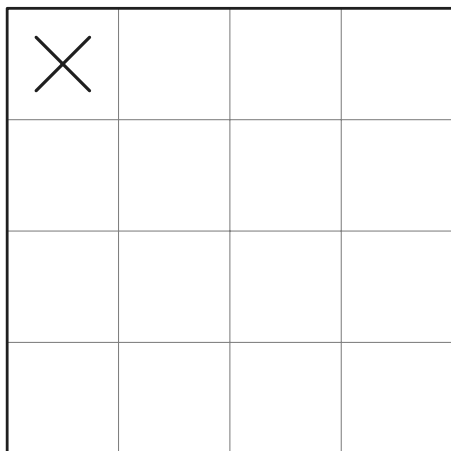
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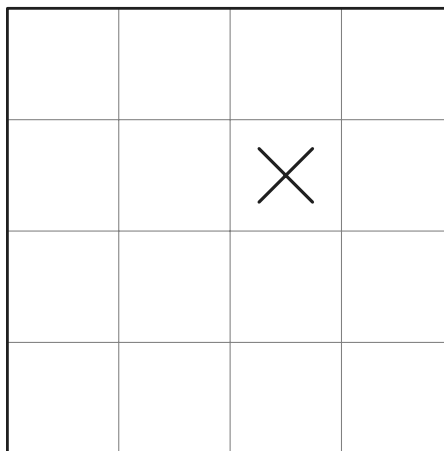
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**Problem 7:** An L-tromino covers three squares of a 2-by-2 block: the block with one square missing. Each board below is 4-by-4 with one square crossed out; on boards (a) and (b) the square is already chosen. Cover each board with 5 paper L-trominoes and draw them. On boards (c) and (d), first cross out a square of your own choice, then cover. Write on the line whether you found a square that makes covering impossible.

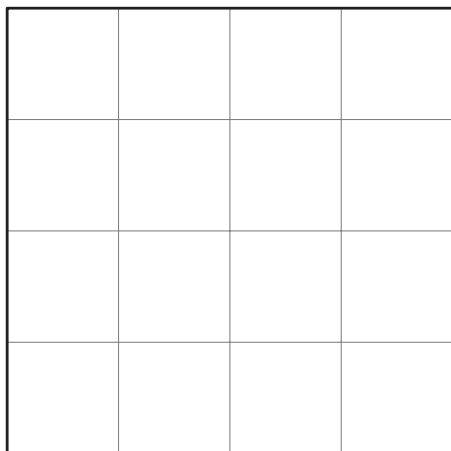
(a)



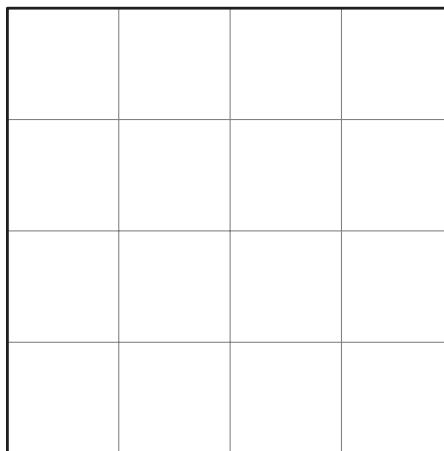
(b)



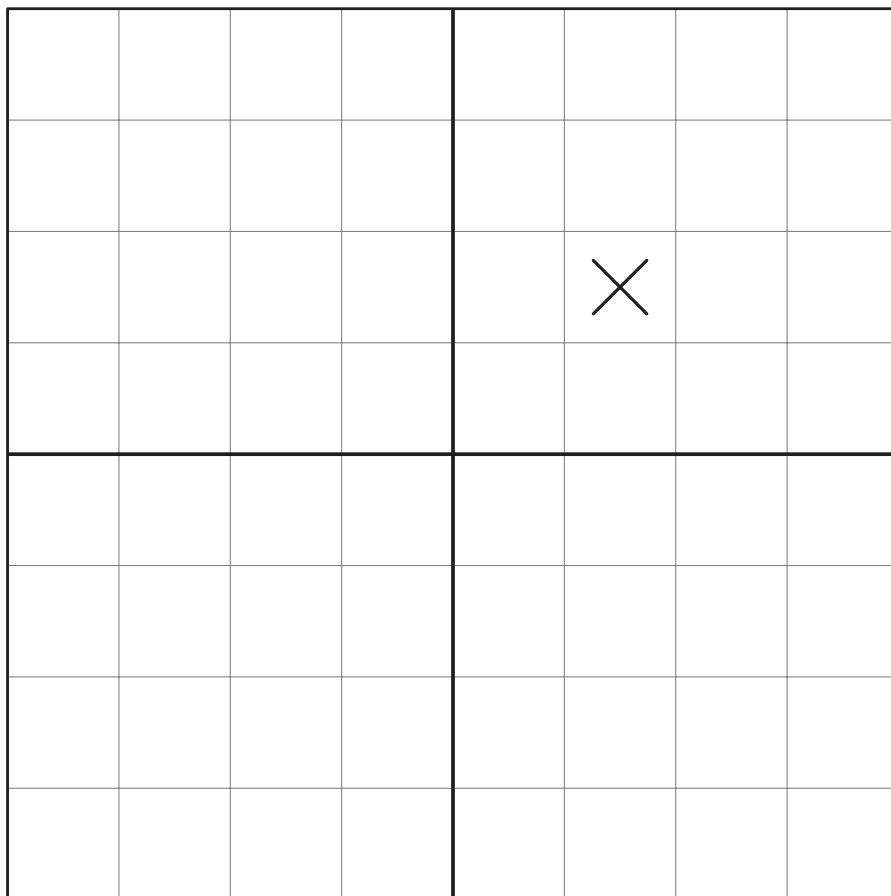
(c)



(d)



**Problem 8:** Board Z is an 8-by-8 board with one square crossed out. Cover it with 21 L-trominoes by this plan, drawing each tromino as soon as you place it so that you can use the paper pieces again. The thick lines split the board into four 4-by-4 quarters. Put one L-tromino in the middle of the board so that it covers one square in each of the three quarters that do not contain the crossed-out square. Now every quarter is a 4-by-4 board with one square missing: repeat the plan inside each quarter, using its four 2-by-2 blocks. Draw all 21 trominoes. Then write on the lines how you would cover a 16-by-16 board with one square crossed out, without doing it, and how many trominoes it would take.

**Z**


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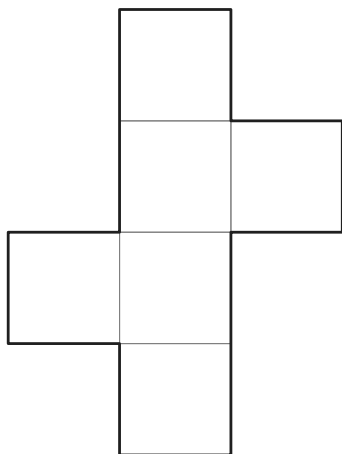
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**Problem 9:** Color board S like a checkerboard and write how many dark and how many light squares it has. Then try to cover it with 3 dominoes. Write  $\checkmark$  or  $\times$ . If  $\times$ , explain on the lines why it is impossible.

**S**



dark:  light:   $\checkmark$  or  $\times$ :

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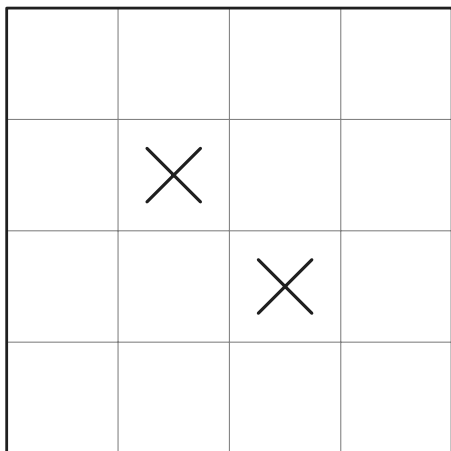
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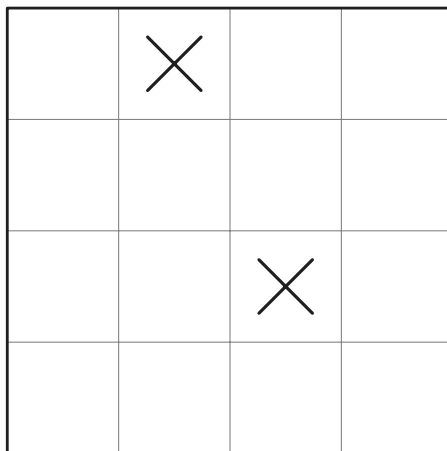
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**Problem 10:** Boards (d), (e) and (f) are 4-by-4 boards with two squares crossed out, like the boards of Problem 5. Do the same: write how many dark and how many light squares are left and your prediction,  $\checkmark$  or  $\times$ , before touching any pieces. Then check with dominoes and write what happened.

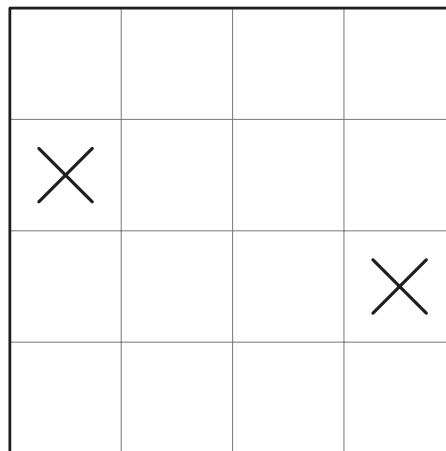
**(d)**



**(e)**



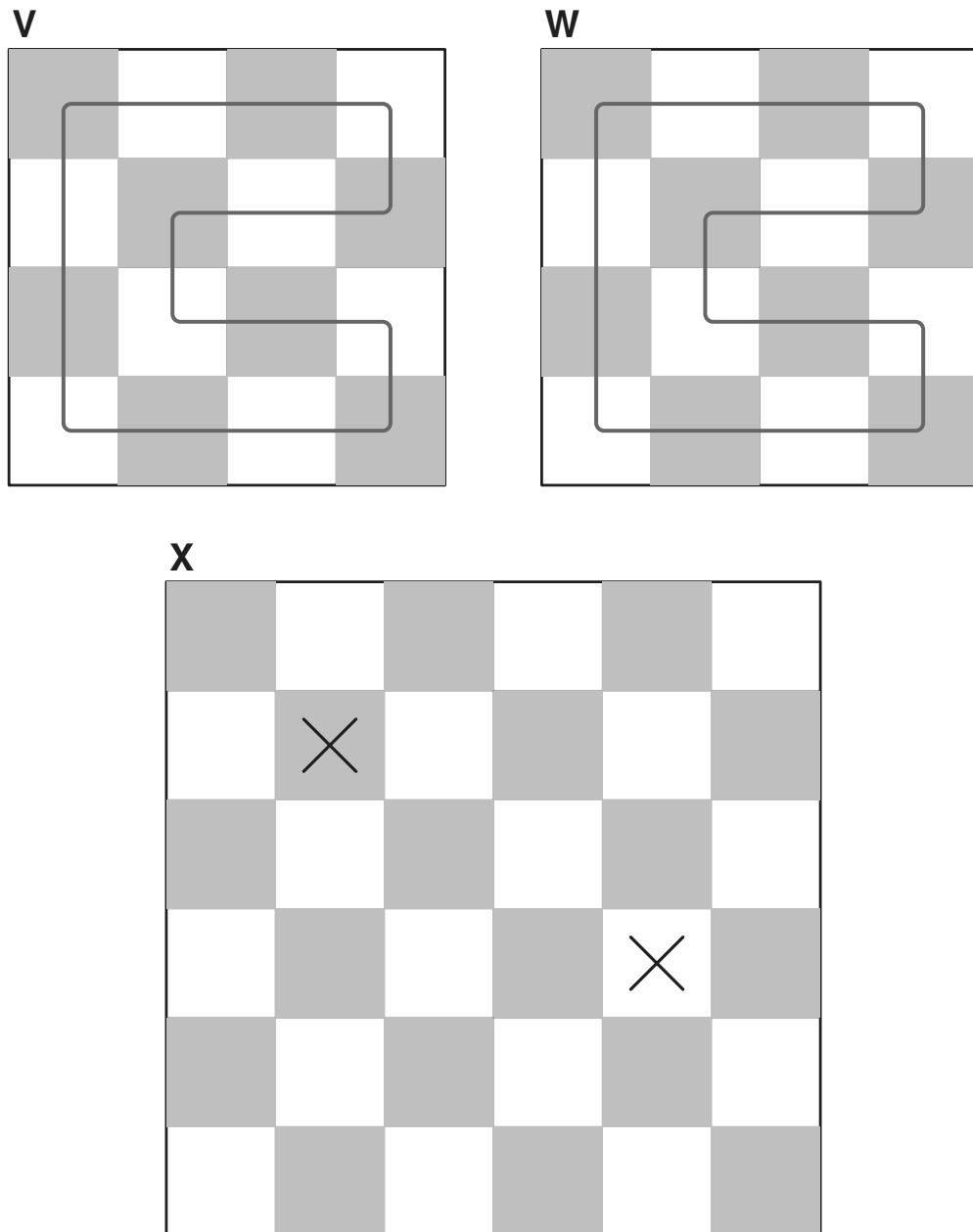
**(f)**



| board | dark left | light left | prediction | result |
|-------|-----------|------------|------------|--------|
| (d)   |           |            |            |        |
| (e)   |           |            |            |        |
| (f)   |           |            |            |        |



**Problem 11:** Board V is a 4-by-4 board with a closed path drawn through all 16 squares. Cross out one dark square and one light square, your choice. Then cover the other 14 squares with paper dominoes by following the path: each domino covers two squares that are next to each other along the path. Draw the dominoes. Do it again on board W with a different pair. Then draw your own closed path through all 36 squares of board X, which already has two squares crossed out, cover board X with dominoes along your path, and draw them. Explain on the lines why the path method always works when one dark and one light square are crossed out.

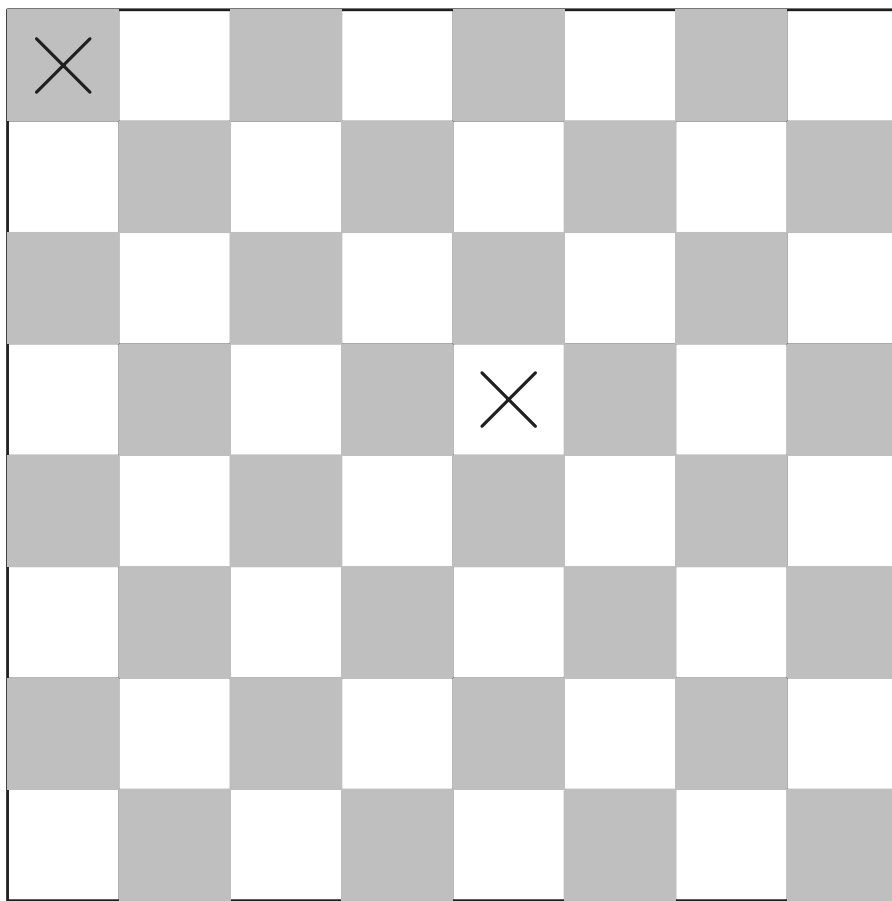


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**Problem 12:** Board Y is an 8-by-8 board with two squares crossed out: the top-left corner and the square in row 4, column 5. Draw a closed path through all 64 squares, ignoring the crosses. Then draw dominoes along your path so that every square except the two crossed-out ones is covered. Then write on the lines whether the path method would work if the two crossed-out squares were both dark, and why.

Y




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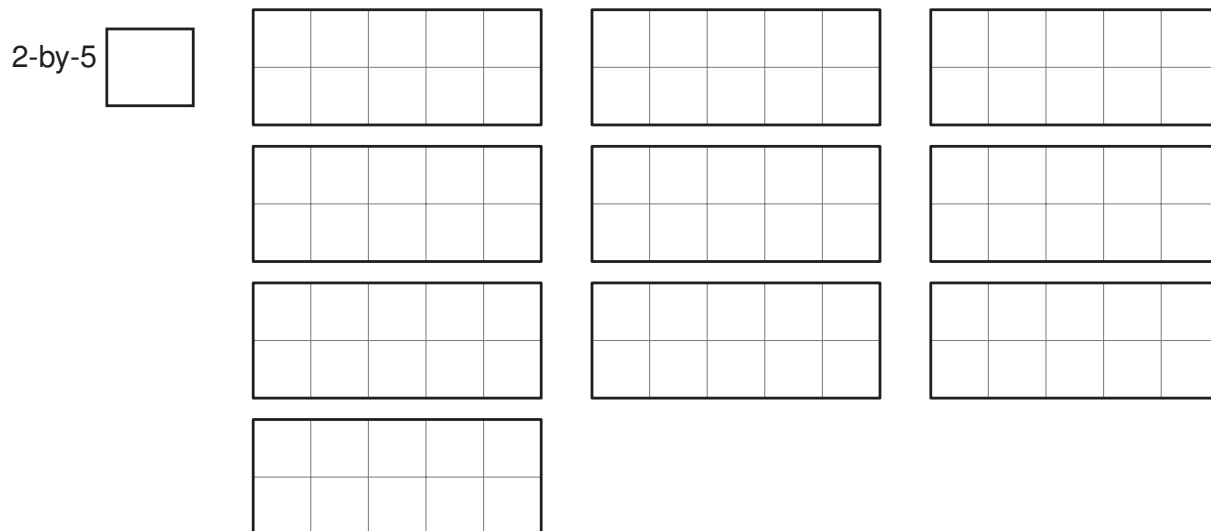
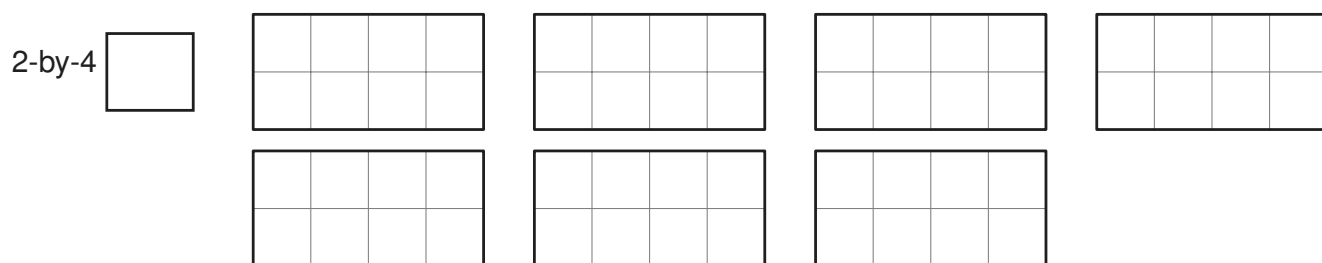
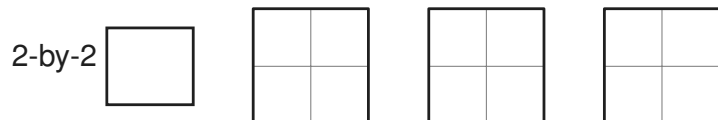
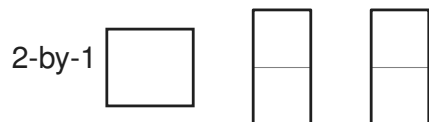


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**Problem 13:** How many different ways are there to cover a 2-by-1 board, a 2-by-2 board, a 2-by-3 board, a 2-by-4 board and a 2-by-5 board with dominoes? Draw every way on the small boards (there are more small boards than you need) and write the number of ways for each size in its box. Then write on the lines your prediction for a 2-by-6 board, and how you can be sure that you found every way for the 2-by-4 board.




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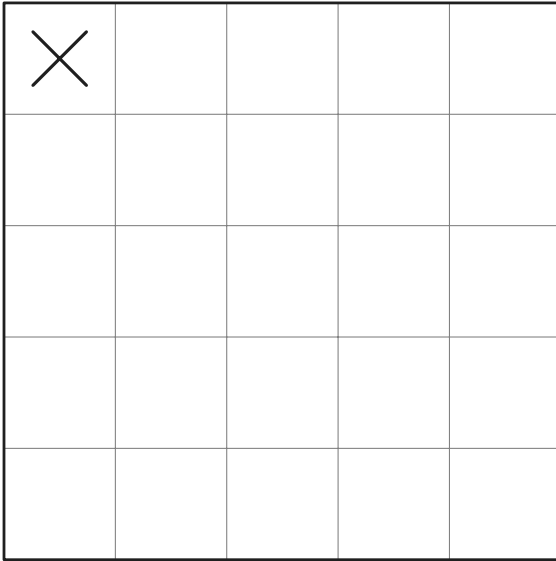
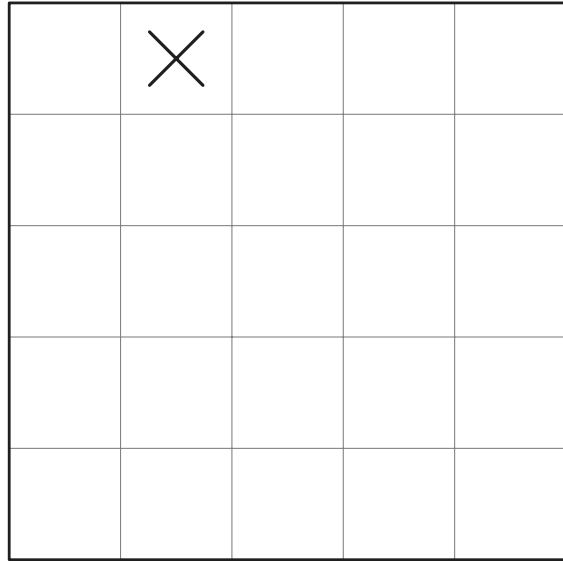


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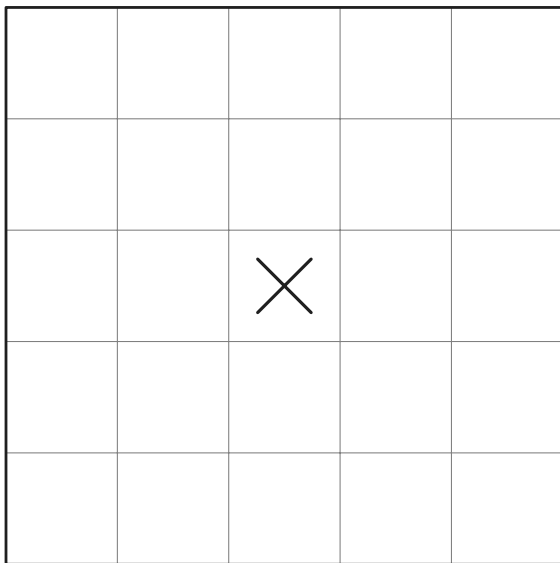
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**Problem 14:** Each board below is 5-by-5 with one square crossed out, so 24 squares are left and 8 L-trominoes would be needed. Cover each board with the paper trominoes if you can, then draw them. Write  $\times$  under the board if you cannot.

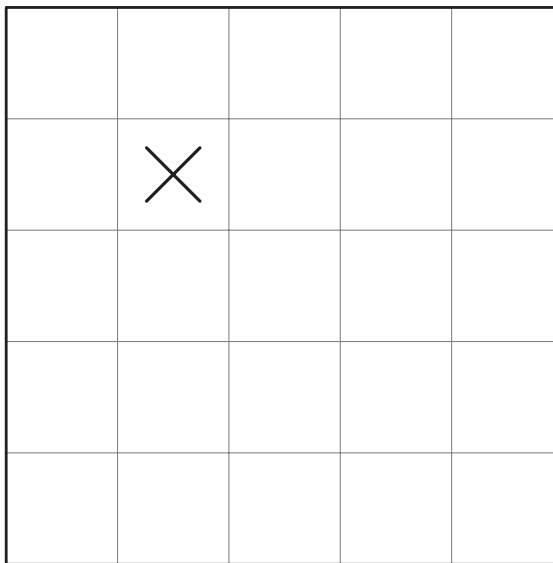
**(a)****(b)**

**Problem 15:** Boards (c) and (d) are 5-by-5 boards with one square crossed out, like the boards of Problem 14. Cover each with the paper trominoes if you can, then draw them; write  $\times$  under the board if you cannot. Then, on the small 5-by-5 map, write  $\checkmark$  in every square that can be crossed out and still leave a board you can cover, and  $\times$  in every square that cannot.

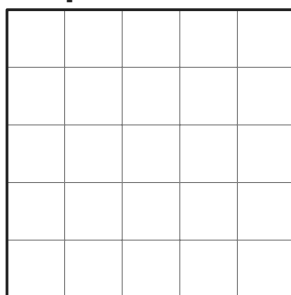
(c)



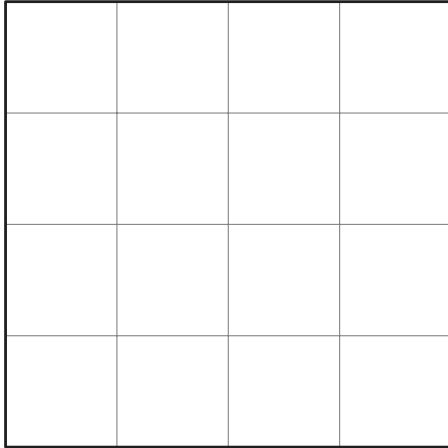
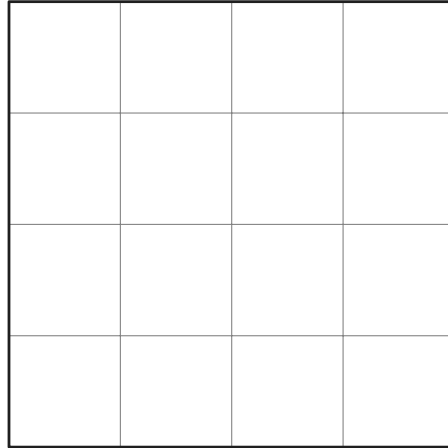
(d)



map



**Problem 16:** Cover board C with exactly 2 vertical dominoes and 6 horizontal ones, and draw it. Cover board D with exactly 4 vertical dominoes and 4 horizontal ones, and draw it. Cover board E with exactly 6 vertical dominoes and 2 horizontal ones, and draw it.

**C****D****E**